ELAVARASI RAVICHANDRAN, Ph.D.

Sensor Data Quality | Bias & Artifact Detection, QC Algorithms & Scientific Data Pipelines

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PROFILE

Atmospheric scientist who spent four years making sure measurements were real before anyone acted on them. Built and owned the quality-control software for five concurrent atmospheric sensors in a federal field campaign — reconciling calibration records against observed performance, tracing anomalous and missing values to their cause, and separating instrument artifact from genuine atmospheric signal. Cross-validated derived quantities against independent co-located instruments run by other institutions, because a sensor that is quietly wrong will agree with itself. Reasons from sensor physics and first principles; published the result as first author in Science Advances. Ph.D. and M.S./B.S. in physics; four years in Julia, comfortable in Python and Linux.

WHAT I BRING

- **Sensor bias and artifact detection —** four years distinguishing instrument malfunction, drift, and contamination from real atmospheric signal across five concurrent sensor streams.
- **Quality-control algorithms —** wrote and maintained the QC code governing which measurements entered analysis and which were flagged or discarded.
- **Independent cross-validation —** verified derived quantities against co-located instruments operated by separate institutions, the only reliable way to catch a silently biased sensor.
- **Calibration and performance verification —** reconciled calibration records against observed behavior; commissioned instruments and verified operating parameters before field deployment.
- **Failure investigation —** traced data loss, gaps, and irregular values to root cause rather than filtering them out.
- **Uncertainty propagation —** carried measurement error through every derived quantity to the reported result.
- **First-principles reasoning —** built a process-scale physical model from governing equations and validated it against observations when published methods did not hold.
- **Cross-team communication —** explained instrument behavior and data issues to four organizations, including a federal agency, who had not made the measurement.

EXPERIENCE

Graduate Researcher — Sensor Data & Quality Control — UC Riverside *Sep 2023 – 2026*

- Owned the processing and quality-control software converting raw output from five concurrent atmospheric sensors into validated scientific data — written in Julia under Linux and in Igor Pro, documented for independent reproduction.
- Reconciled calibration records against observed instrument performance and investigated every anomalous, irregular, and missing value to root cause, deciding in each case whether it was instrument artifact or real atmospheric signal.
- Cross-validated derived quantities against independent co-located instruments operated by other institutions on the same platform, catching discrepancies that single-instrument checks would not surface.
- Propagated measurement uncertainty through every derived quantity to the final reported result.
- Found that removing atmospheric trace gases can either increase or decrease cloud droplet formation depending on conditions — overturning an assumed one-directional relationship. First author, Science Advances (2026).
- Built a process-scale evaporation model from first principles and validated it against the observational record when existing published methods proved insufficient.
- Served as the technical point of contact across four organizations — Environment and Climate Change Canada, Scripps Institution of Oceanography, a national field campaign, and UC Riverside — reconciling data conventions and resolving measurement disputes.

Research Assistant — Instrument Commissioning — NC State University *Aug 2022 – Jul 2023*

- Assembled, bench-tested, and commissioned the full atmospheric sensor suite for a Department of Energy field campaign, verifying each instrument against expected operating parameters and documenting results before shipment.
- Troubleshot hardware faults during preparation and resolved them before deployment, where diagnosis would have been far harder.
- Performed first-pass processing and quality review of returning field data.

Research Assistant — Atmospheric Measurement & Modeling — SRM Institute, Chennai *2017 – 2020*

- Conducted ambient particulate field sampling and laboratory analysis; quantified concentrations and emission rates and apportioned them among sources; co-authored the resulting publication.
- Ran the NOAA HYSPLIT transport model on gridded meteorological reanalysis across a 15-year, seven-region dataset.

Research Intern, Space Physics Laboratory — Indian Space Research Organisation (ISRO) *May 2019*

EDUCATION

Ph.D., Chemical and Environmental Engineering — University of California, Riverside *Sep 2026*

M.S., Physics — SRM Institute of Science and Technology, Chennai, India *2020*

B.S., Physics — SRM Institute of Science and Technology, Chennai, India *2018*

TECHNICAL SKILLS

Programming: Julia (4 years, primary); Python; R; Igor Pro; Linux, macOS, Windows

Sensor Data: Quality-control algorithm development; bias, drift, and artifact detection; calibration record review and performance verification; failure and root-cause investigation; cross-validation against independent instruments; uncertainty quantification and error propagation

Instrumentation: Scanning Mobility Particle Sizer (SMPS); Single Particle Soot Photometer (SP2); Cloud Condensation Nuclei Counter (CCNC); Humidified Tandem Differential Mobility Analyzer (HTDMA); Printed Optical Particle Spectrometer (POPS); thermal-optical OC/EC carbon analyser; instrument rack assembly, bench testing, and commissioning

Science: Atmospheric physics and thermodynamics; cloud microphysics; heat and mass transfer; statistical and trend analysis; process-scale model development; NOAA HYSPLIT transport modeling